



OUR LADY OF FATIMA UNIVERSITY

College of Computer Studies – Quezon City

Final Project

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Course/Yr/Sec.:		BSIT 1 YB-8	

Project Title: Flight Booking System

Project Description:

The flight booking system is an innovative, text-based platform designed to make travel planning easier. It allows users to effortlessly book flights through a simple, user-friendly interface. The system offers personalized options, allowing users to specify preferences such as domestic or international flights, dates, times, and seat classes, and seat slot. It provides real-time updates on availability and ensures a smooth booking experience tailored to individual needs. This system offers great control and flexibility over travel plans, making the process efficient and enjoyable for every traveler. With a strong focus on user experience, it makes travel management convenient and straightforward

Below are the capabilities and limitations of the proposed project:

Capabilities:

The flight booking system allows users to book, create accounts, and manage flight bookings through text commands. This systems provide functionalities such as:

User Account Management:

- Create and administer personal accounts
- Manage and update personal information
- Secure login and authentication
- Account Retrieval

Customized Booking Options:

- Select seat class
- Choose preferred flight times and dates
- Options for both domestic and international flights
- Tailor accommodations to match user schedules

Booking Management:

- Cancel and modify bookings through the user menu
- Real-time availability updates for seamless booking experience
- View and manage past and upcoming bookings

User-Friendly Interface:

- Intuitive and easy-to-navigate design

- Clear and concise prompts to guide users through the booking process
- Simple and safe access to all booking features

Administrator Dashboard:

- Comprehensive management of user accounts
- Oversee and update available flight listings
- Monitor and manage overall booking activity

Priority on User Satisfaction:

- Enhanced user experience through efficient booking workflows
- Reliable and secure system for user confidence and trust

Limitations:

Amidst the pursuit of innovation, certain limitations persist, potentially impacting user satisfaction and usability. Understanding these limitations is crucial for both developers and users alike, as they shape the functionality and accessibility of travel platforms.

Limited customization options, despite offering tailored options, the level of customization may be limited, preventing users from fully personalizing their travel experiences according to their preferences and requirements.

Integrating with external systems, such as airline reservation systems, can pose challenges. Potentially affect the reliability and accuracy of booking information.

Language support, the system is limited to English; users from different locations or linguistic backgrounds may find it difficult to navigate.

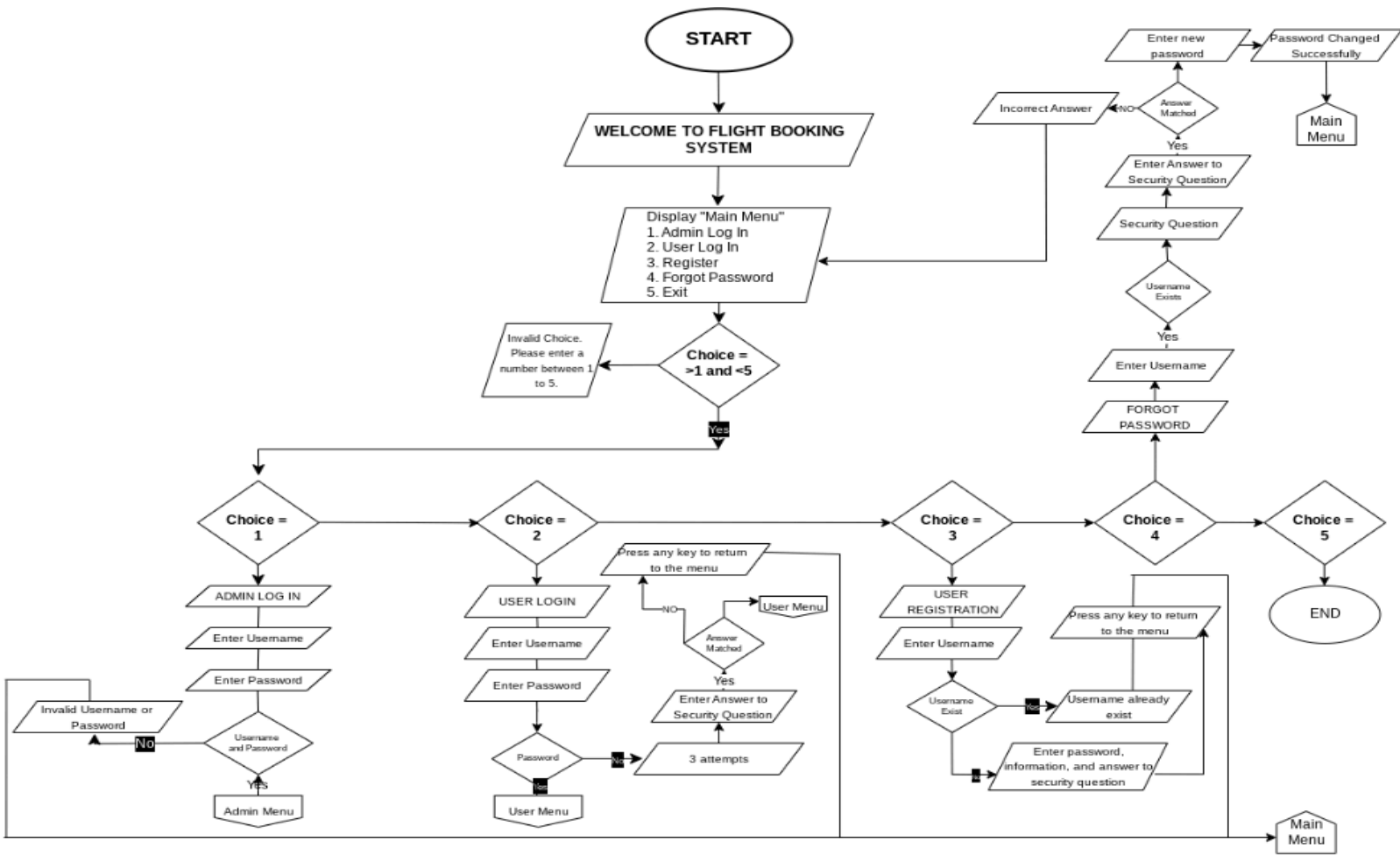
In **less-traveled areas**, the platform may have limited availability of accommodations and booking options, which may not be ideal for travelers seeking off-beaten-path destinations. The available flights will only depend on the admin's update.

The **text-based interface** of the system may present difficulties for users who are accustomed to user interfaces (GUIs). It lacks cues. It requires users to navigate through text menus and commands, which could result in a steeper learning curve.

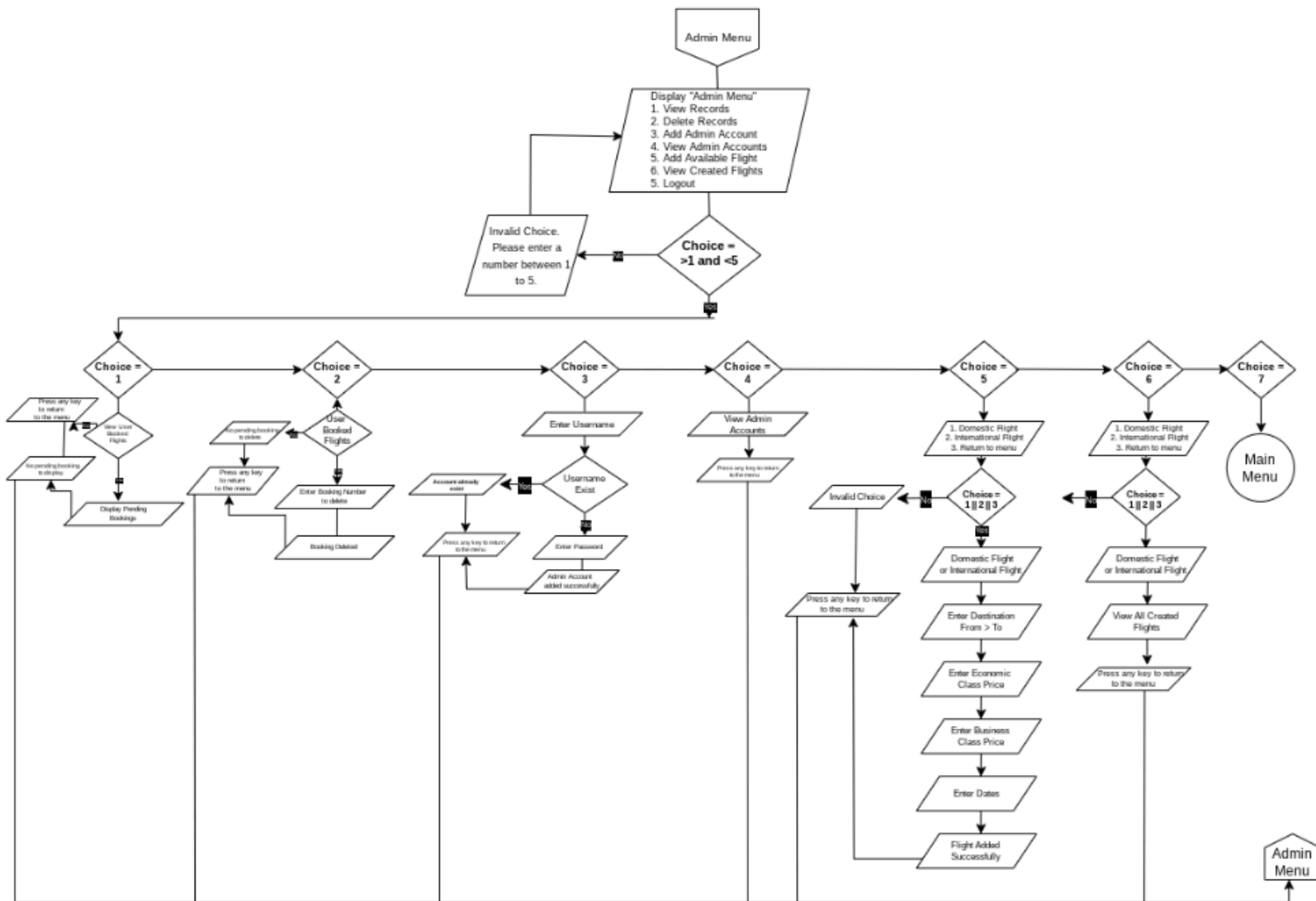
Booking Limitation. The user cannot book if there's no available flight updated by the admin.

Project Diagram

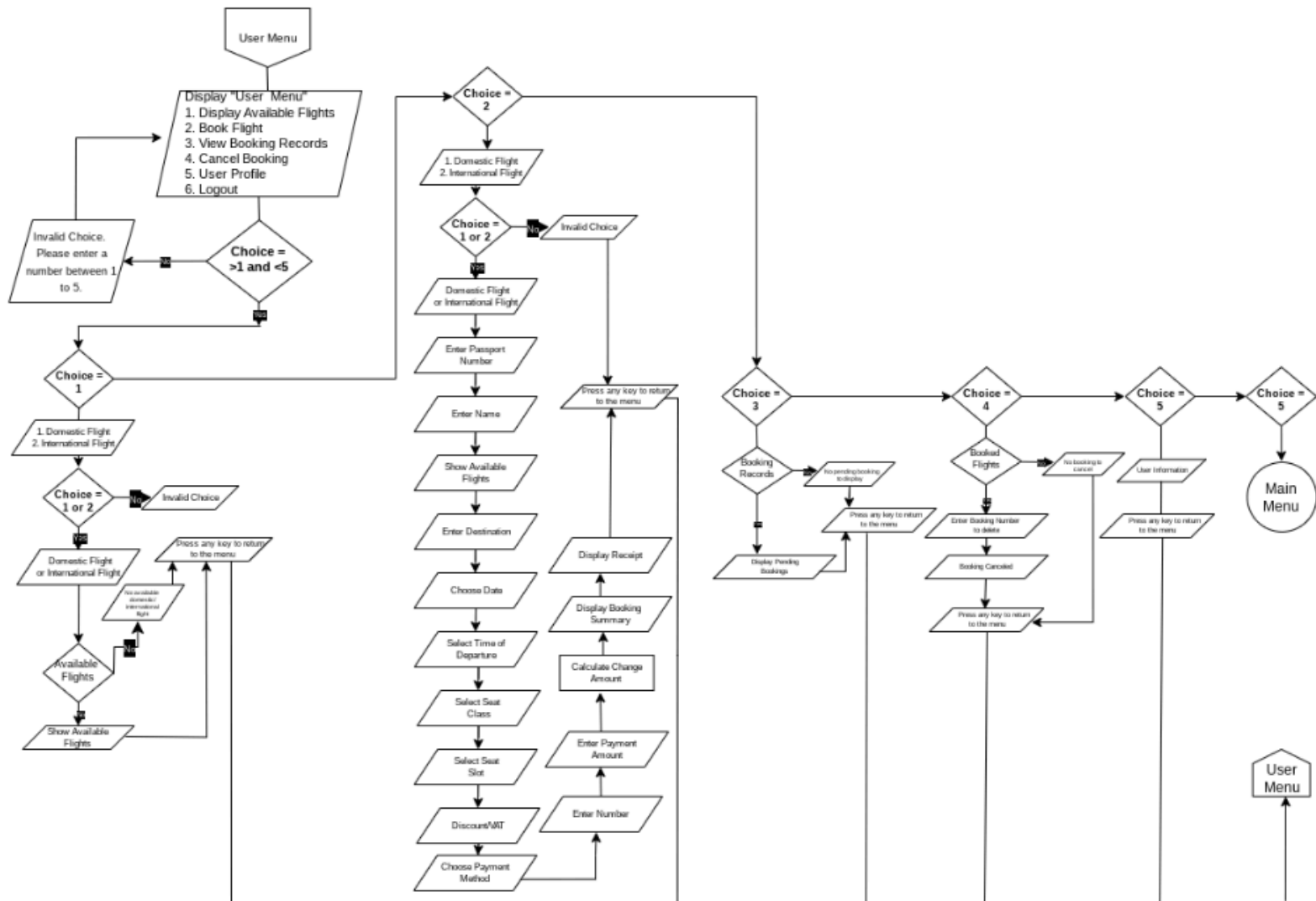
START



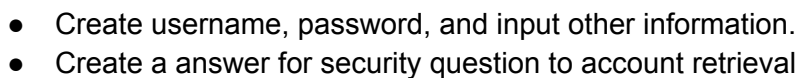
ADMIN MENU



USER MENU



Main Menu



Login as User



3 Attempts to login if forgot password

```
----- User Log In -----  
  
Enter username: bitpair  
Enter password: *****  
Invalid password! Try again.  
Enter password: ****  
Invalid password! Try again.  
Enter password: ***  
Invalid password! Try again.  
  
You've exceeded the maximum number of login attempts.  
Answer the security question to continue.  
Security Question: What's your favorite color?  
  
Your Answer: red|
```

- Added security question for account retrieval

User Menu

```
+-----+  
|           User Menu           |  
+-----+  
| 1. Display Available Flights |  
| 2. Book Flight               |  
| 3. View Booking Records     |  
| 4. Cancel Booking           |  
| 5. View Profile             |  
| 6. Log Out                  |  
+-----+  
  
Enter your choice: |
```

View available flights

```
----- Available Flights -----  
  
1. Domestic Flights  
2. International Flights  
3. Return to Menu  
  
Enter your choice: |
```

- Choose domestic or international flight
- Added Option to return to menu

```

----- Available Flights -----

Destination: manila to cebu
Economic Class Price: 900.00
Business Class Price: 1000.00
Available Dates: 08/14/24 08/15/24

Destination: manila to davao
Economic Class Price: 1200.00
Business Class Price: 1500.00
Available Dates: 09/12/24 09/13/24 09/14/24

Press any key to return to Menu.

```

- See all available flights to book

Booking a flight

```

----- BOOK FLIGHT -----

Choose Flight:
1. Domestic Flight (Local Flight)
2. International Flight
3. Return to Menu
Enter your choice: |

```

- Choose flight to book

```

***** Booking Details *****

Enter Passport Number: 12345
Enter Name: bitpair

Available Domestic Flights:
+-----+
| Destination | Econo | Business |
+-----+
| 1. manila to cebu | P900.00 | P1000.00 |
| 2. manila to davao | P1200.00 | P1500.00 |
+-----+

Enter Destination: 1
Available Dates:
1. 08/14/24
2. 08/15/24
Select Date: 2

+-----+
| Time of Departure |
+-----+
| 1. 4:00am | 2. 7:00am | 3. 9:00am |
| 4. 5:00pm | 5. 7:00pm | 6. 9:00pm |
+-----+

Select Time: 3
Select Seat Class:
1. Economic
2. Business
Enter your choice: 1|

```

- Enter booking details

```
***** Select Seat *****

Seat Map (0 means available, 1 means booked):
  0 0 0 0 0 0 0 0 0 0
  0 0 0 0 0 0 0 0 0 0

  0 0 0 0 0 0 0 0 0 0
  0 0 0 0 0 0 0 0 0 0

Select a seat (0-39): 23|
```

- Select seat slot (added)

```
***** Payment *****

Are you eligible for a 20% discount for elders or PWDs? (Y/N): y

Select Payment Method:
1. Credit Card
2. GCash
3. PayMaya

Enter your choice: 1
Enter Card/Gcash/Paymaya Number: 123456
Enter Payment Amount: 2000|
```

- Eligibility for discount and add VAT
- Select Payment Method and Payment Amount

```
***** Booking Summary *****

+-----+
| Booking Details |
+-----+
| Attribute | Value |
+-----+
| Booking Number | 1 |
| Passport Number | 12345 |
| Name | bitpair |
| Flight Class | Economic |
| Destination | manila to cebu |
| Date | 08/15/24 |
| Time | 9:00am |
| Seat Number | 23 |
+-----+

Booking Successful!
Do you want to print the receipt? (Y/N): y|
```

- Print booking summary and ask to print receipt

RECEIPT	
Attribute	Value
Ticket Price	900.00
Discount/VAT Applied	180.00
Total Payable Amount	720.00
Payment Type	Credit Card
Card Number	123456
Paid Amount	2000.00
Change	1280.00

Press any key to return to User Menu.

- Print detailed receipt with discount/VAT

Viewing all booking records (if any)

```

----- Booking Records -----
Booking Records:
Booking Number: 1
Passport Number: 12345
Name: bitpair
Flight Class: Economic
Destination: manila to cebu
Date: 08/15/24
Time: 9:00am
Seat Number: 23
Price: 900.00

Booking Number: 2
Passport Number: 213231
Name: bitpair
Flight Class: Business
Destination: manila to davao
Date: 09/14/24
Time: 7:00pm
Seat Number: 15
Price: 1500.00

Press any key to return to User Menu.

```

- View all booking records

Cancel Booking

```
----- Cancel Booking -----  
Enter Booking Number to Cancel: 1  
Are you sure you want to cancel booking 1? (y/n): n  
Booking cancellation aborted.  
  
Press any key to return to User Menu.
```

- Added confirmation question

```
----- Cancel Booking -----  
Enter Booking Number to Cancel: 1  
Are you sure you want to cancel booking 1? (y/n): y  
Booking 1 canceled successfully.  
  
Press any key to return to User Menu.
```

```
----- Booking Records -----  
  
Booking Records:  
  
Booking Number: 2  
Passport Number: 213231  
Name: bitpair  
Flight Class: Business  
Destination: manila to davao  
Date: 09/14/24  
Time: 7:00pm  
Seat Number: 15  
Price: 1500.00  
  
Press any key to return to User Menu.
```

- Cancel Booking
- (Booking #1 was already canceled)

```
----- User Profile -----  
  
User Profile  
Username: bitpair  
Name: BitPair  
Age: 19  
Address: qc  
Gender: f  
  
Press any key to return to User Menu.
```

- Added user profile for user information

Logout User Menu

```
+-----+
|               |
|      User Menu      |
|               |
| 1. Display Available Flights |
| 2. Book Flight       |
| 3. View Booking Records |
| 4. Cancel Booking    |
| 5. View Profile      |
| 6. Log Out           |
|               |
+-----+

Enter your choice: 6
Are you sure to log out account? (1 for Yes, 0 for No): 1|
```

- Added Confirmation prompt to logout user account

Login as Admin

```
----- Admin Log In -----

Enter admin username: Admin
Enter admin password: *****|
```

- Enter default admin account

Admin Menu

```
+-----+
|               |
|      Admin Menu      |
|               |
| 1. View Records      |
| 2. Cancel User Booking |
| 3. Create Admin Account |
| 4. View Admin Accounts |
| 5. Add Available Flights |
| 6. View Created Flights |
| 7. Log out           |
|               |
+-----+

Enter your choice: |
```

View record of bookings (if any)

```
----- Booking Records -----  
  
Booking 2  
Passport Number: 213231  
Name: bitpair  
Flight Class: Business  
Destination: manila to davao  
Time: 7:00pm  
Price: 1500.00  
  
Press any key to return to Admin Menu.
```

- Display all booking user records (if any)

Cancel user booking

```
----- Cancel User Booking -----  
  
Enter Booking Number to Cancel: 2  
Are you sure you want to cancel booking 2? (y/n): y  
Booking 2 cancelled successfully.  
Press any key to return to Admin Menu.
```

- Added confirmation prompt to cancel user booking

Adding new admin account

```
----- Add Admin Account -----  
  
Enter admin username: admin1  
Enter admin password: 12345  
Admin account added successfully!  
Press any key to return to Admin Menu.
```

- Create new admin username and password

Admin Profile/View Admin Accounts (added)

```
----- Admin Accounts -----  
  
Admin Account 1  
Username: admin  
  
Admin Account 2  
Username: adm  
  
Admin Account 3  
Username: admin1  
  
Press any key to return to Admin Menu.
```

Adding new available flight

```
----- Add Flight -----

1. Domestic Flight
2. International Flight
3. Return to Menu

Choice: 1

Add Domestic Flight:

Destination (From > To): manila to palawan
Economic Class Price: 800
Business Class Price: 1000
Enter number of available dates: 3
Enter available date 1 [MM/DD/YY]: 07/01/24
Enter available date 2 [MM/DD/YY]: 07/02/24
Enter available date 3 [MM/DD/YY]: 07/03/24

Flight added successfully!

Press any key to return to Admin Menu.
```

- Choose either domestic or international flight to add
- Added option to return to menu
- Enter destination, economic price, business price and added dates for flight

View Created Flights (Added)

```
----- Available Flights -----

Destination: manila to cebu
Economic Class Price: 900.00
Business Class Price: 1000.00
Available Dates: 08/14/24 08/15/24

Destination: manila to davao
Economic Class Price: 1200.00
Business Class Price: 1500.00
Available Dates: 09/12/24 09/13/24 09/14/24

Destination: manila to palawan
Economic Class Price: 800.00
Business Class Price: 1000.00
Available Dates: 07/01/24 07/02/24 07/03/24

Press any key to return to Menu.
```

Logout Admin Account

```
+-----+
|           Admin Menu           |
+-----+
| 1. View Records                |
| 2. Cancel User Booking         |
| 3. Create Admin Account        |
| 4. View Admin Accounts         |
| 5. Add Available Flights       |
| 6. View Created Flights        |
| 7. Log out                     |
+-----+

Enter your choice: 7
Are you sure to logout admin account? (1 for Yes, 0 for No): 1|
```

- Added confirmation prompt to logout admin account

Forgot password for account retrieval (added in main menu)

```
----- Forgot Password -----

Enter your username: bitpair

Security Question: What is your favorite color?
Provide the answer to the security question: red

Security answer matched!
Enter new password: *****
Password changed successfully!

Press any key to return to Main Menu.
```

- Answer to security question should match or valid to create a new password